

1	Course Name:
	AI PROGRAMMING
2	Course Code:
	IQB10104
3	Year and Semester Offered:
	Year 1 Semester 1
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	NA
6	Course Synopsis:
	This course introduces students to Python programming as a foundation for AI development. It focuses on core programming concepts, problem-solving techniques, and Object-Oriented Programming (OOP) principles. Students will learn essential programming constructs, data structures, and algorithmic thinking and applying these concepts to AI-related problem-solving.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Express an understanding of fundamental programming concepts and Object-Oriented Programming (OOP) principles using Python. 2. Apply the object-oriented programming principles into written Python Codes by reusing existing library classes. 3. Builds object-oriented programs or applications using Python for simple AI based solutions.
8	Content Outline:
	Topic
	1. Introduction to Python and AI
	2. Object and Basic Data Structure
	3. Control Structures
	4. Advanced Data Structure : Lists, Tuples, Dictionaries and Sets
	5. Modular Programming: Methods and Functions
	6. Object-Oriented Programming (OOP) Basics
	7. OOP Concepts: Inheritance & Polymorphism
	8. Errors and Exception Handling
	9. Modules and Packages, Built-in functions
	10. Algorithmic Thinking for AI

	11. Basic AI-related Programming
	12. Debugging and Code Optimization
9	References:
	T. Gaddis, Starting Out with Python, Pearson, 6th edition (2022). [ISBN-13: 9780137871209]